

Pressure-to-Deflection Modelling of an Eccentric-Cavity Soft Pneumatic Bending Actuator

B.Tech Project — progress review

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Where the project stands

Three of the five stages are complete; multi-cavity designs and ANSYS verification come next

✓ DONE

1

Derivation

Derived the section properties, pressure loading and bending, step by step.

✓ DONE

2

Analytical models

Three beam-model tiers: linear, (1–2v) factor, wall expansion.

✓ DONE

3

3-D nonlinear FEA

In-house hyperelastic FE solver, verified, then swept the bare and sleeved designs.

NEXT · 5–25 OCT

4

Multi-cavity + ANSYS

Several eccentric cavities for bending in any x–y direction, then an ANSYS check of every model.

FROM 26 OCT

5

Mould + tests

3-D printed mould, cast actuators, pressure tests compared with the models.

IN THIS TALK

A · The derivation, step by step

B · Key results and model tiers

C · FEA: how we built it

D · FEA: what we found

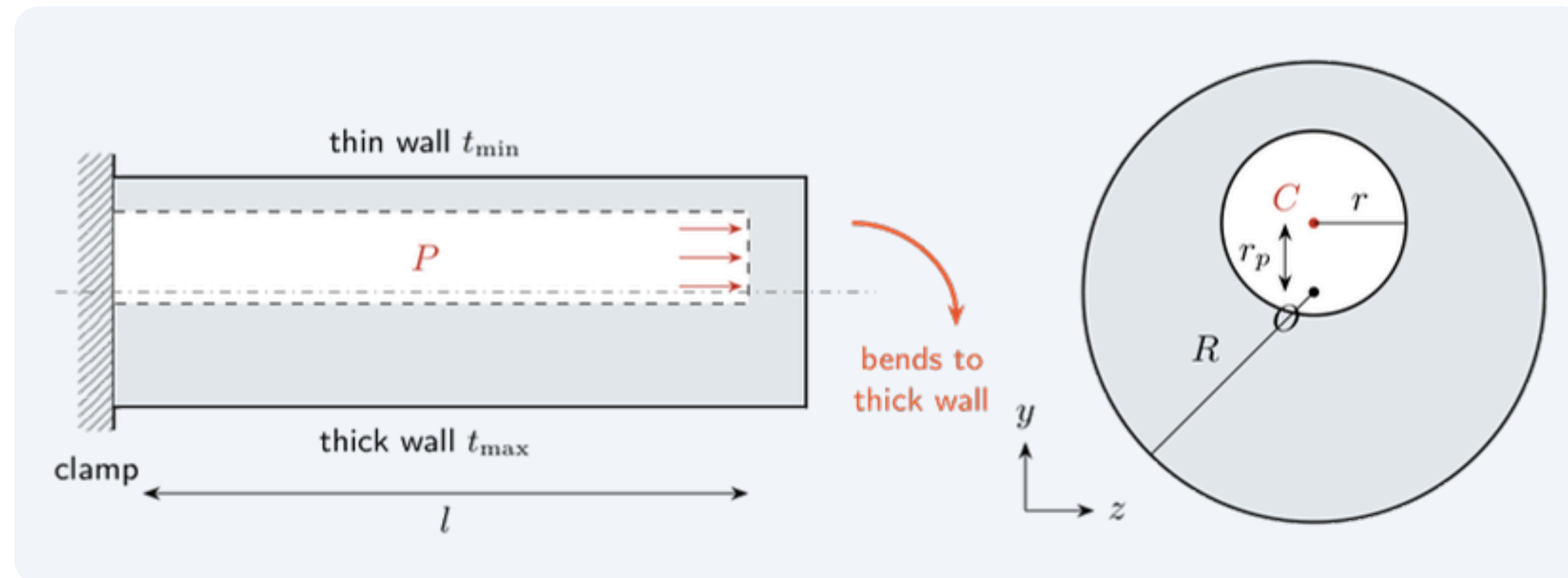
E · Plan for the next month

HEADLINE RESULT SO FAR

The bare actuator balloons and snaps through at 104.7 kPa. A hoop-reinforced actuator bends smoothly, 2.8× more at 60 kPa, and the simple beam models match it within 5 % at low pressure. We recommend building the reinforced design.

An off-centre cavity turns pressure into bending

Goal: a reliable map from cavity pressure P to tip deflection δ , first by hand, then with 3-D FE, then by experiment



PLACEHOLDER DESIGN

Outer radius $R = 10$ mm

Cavity radius $r = 4$ mm

Cavity offset $r_p = 3$ mm

Length $L = 100$ mm

Thin wall 3 mm

Thick wall 9 mm

Silicone neo-Hookean

$E = 0.6$ MPa ($\mu = 0.2$ MPa)

Pressure 0 – 110 kPa

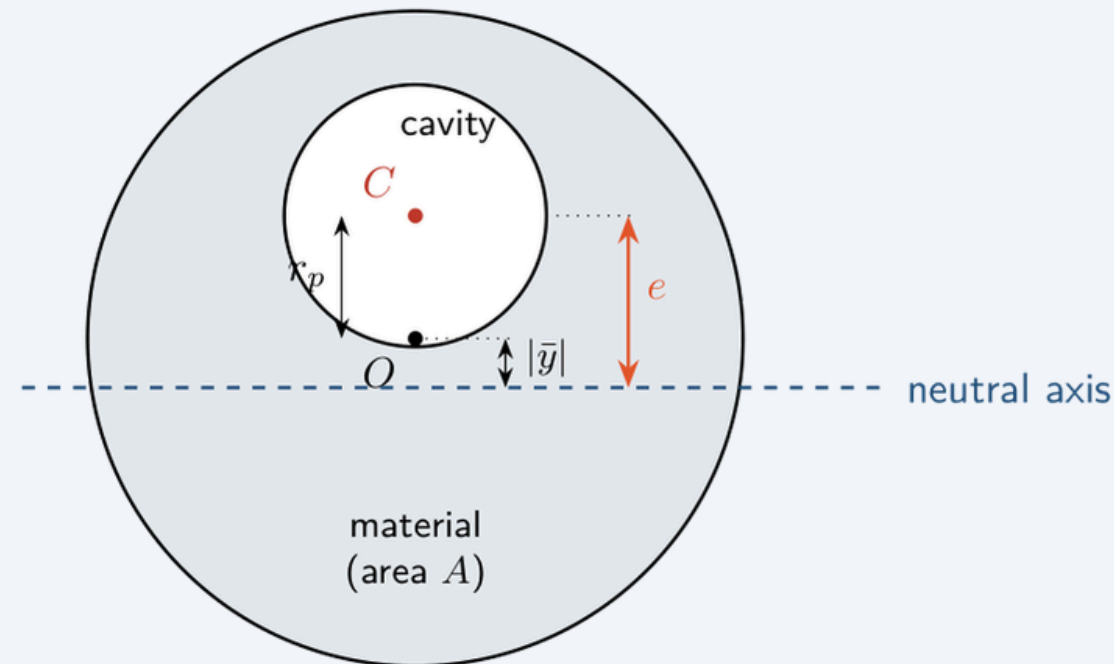
These are placeholder values. Every model in this deck is parametric, so the real geometry and silicone drop straight in.

HOW IT BENDS

Air pressure P pushes on the cavity wall. Because the cavity sits 3 mm off-centre, the wall on one side is thin (3 mm) and on the other thick (9 mm). The thin side stretches more, so the tube curls towards the thick side. The question is how far the tip moves for a given pressure, and which model predicts it reliably.

Cross-section: area and neutral axis

What we did: located the neutral axis of the off-centre (eccentric) section



not to scale: $\bar{y}$ exaggerated

The cavity is a negative area centred at C, so the neutral axis moves toward the thick wall.

FINDING

The neutral axis sits 0.57 mm below the outer centre, on the thick-wall side: $\bar{y} = -r^2 r_p / (R^2 - r^2)$.

HOW — STEP BY STEP

- 1 Treat the section as the outer disc minus the cavity disc.

$$A = \pi R^2 - \pi r^2 = \pi (R^2 - r^2)$$

- 2 The centroid of a composite area is $\sum A_i y_i / \sum A_i$. The outer disc is centred at $y = 0$; the cavity is a *negative* area centred at $y = r_p$.

$$\bar{y} = \frac{\pi R^2 \cdot 0 - \pi r^2 \cdot r_p}{\pi R^2 - \pi r^2}$$

- 3 Cancel π . Material is missing on the $+y$ side, so the neutral axis moves toward the thick wall ($\bar{y} < 0$).

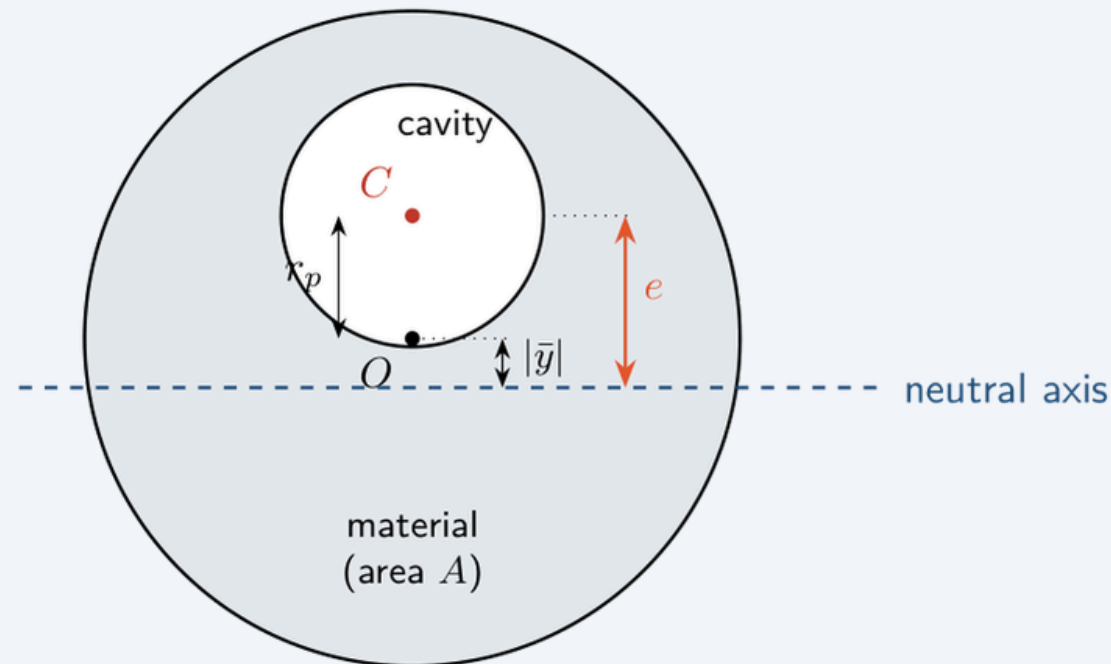
$$\bar{y} = -\frac{r^2 r_p}{R^2 - r^2}$$

- 4 Placeholder design ($R = 10$, $r = 4$, $r_p = 3$ mm):

$$A = 263.9 \text{ mm}^2, \quad \bar{y} = -0.571 \text{ mm}$$

Cross-section: second moment of area

What we did: computed the bending stiffness I_{zz} of the eccentric section



not to scale: $\bar{y}$ exaggerated

Parallel-axis theorem about the neutral axis: outer disc at $|\bar{y}|$,
cavity at $e = r_p - \bar{y}$.

FINDING

$I_{zz} = 7114 \text{ mm}^4$, 7 % below a concentric tube (7653 mm^4). The identity $r_p - \bar{y} = R^2 r_p / (R^2 - r^2)$ is reused for the moment arm.

HOW — STEP BY STEP

- 1 Parallel-axis theorem, $I = I_c + A d^2$, for each disc about the neutral axis. Distances: outer disc $d = |\bar{y}|$, cavity $d = r_p - \bar{y}$.

$$I_{zz} = \left[\frac{\pi R^4}{4} + \pi R^2 \bar{y}^2 \right] - \left[\frac{\pi r^4}{4} + \pi r^2 (r_p - \bar{y})^2 \right]$$

- 2 A shortcut we use again later: the cavity distance simplifies to

$$r_p - \bar{y} = r_p + \frac{r^2 r_p}{R^2 - r^2} = \frac{R^2 r_p}{R^2 - r^2}$$

- 3 Substitute $\bar{y}$ and $r_p - \bar{y}$ into the two $A d^2$ terms and factor:

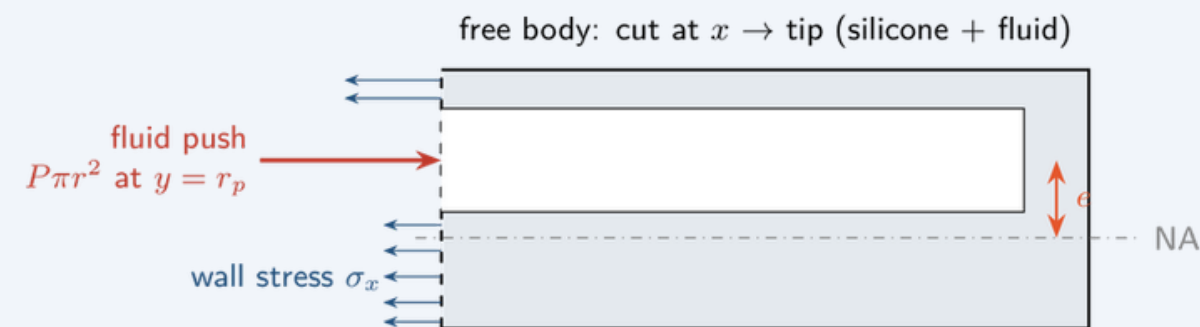
$$\pi R^2 \bar{y}^2 - \pi r^2 (r_p - \bar{y})^2 = \frac{\pi r_p^2 (R^2 r^4 - r^2 R^4)}{(R^2 - r^2)^2} = - \frac{\pi R^2 r^2 r_p^2}{R^2 - r^2}$$

- 4 Result: concentric hollow tube, minus the stiffness lost by moving the hole off-centre.

$$I_{zz} = \frac{\pi (R^4 - r^4)}{4} - \frac{\pi R^2 r^2 r_p^2}{R^2 - r^2} = 7652.9 - 538.6 = 7114 \text{ mm}^4$$

Pressure load and bending moment

What we did: turned the cavity pressure into an end thrust and the bending moment it creates



Cut the actuator at any x and keep the tip piece together with the fluid inside it.

The fluid beyond the cut pushes with $P\pi r^2$ through the cavity centre C ; the silicone wall pulls back with σ_x .

The push acts a distance e above the neutral axis, so it bends the actuator with $M = F \cdot e$.

FINDING

$M = F \cdot e$ is linear in P and the same at every cross-section (0.18 N·mm per kPa). The thrust acts on the cavity area: $F = P \cdot \pi r^2$.

HOW — STEP BY STEP

- 1 Pressure acts on the inside face of the sealed tip, whose area is the cavity area $A_c = \pi r^2$:

$$F = P A_c = P \pi r^2$$

- 2 A uniform pressure on a circle has its resultant at the centre of the circle, $y = r_p$. Its lever arm about the neutral axis (step 2 of the previous slide):

$$e = r_p - \bar{y} = \frac{R^2 r_p}{R^2 - r^2}$$

- 3 Moment = force $\times$ lever arm:

$$M = F e = \frac{\pi P R^2 r^2 r_p}{R^2 - r^2}$$

- 4 Why M is the same everywhere: cut at any x , keep the tip piece *with its fluid*. Only the cut carries load, so

$$\int_A \sigma_x dA = P \pi r^2, \quad \int_A \sigma_x (y - r_p) dA = 0$$

- 5 Numbers: per kPa of pressure $F = 0.050$ N and $M = 0.180$ N mm.

Beam response: curvature, tip angle, deflection

What we did: converted the moment into curvature, tip angle and tip deflection (Euler–Bernoulli beam)

HOW — STEP BY STEP

- 1 Euler–Bernoulli: plane sections stay plane and every fibre is in uniaxial stress, $\sigma_x = E [\varepsilon_0 + \kappa (y - \bar{y})]$. Taking moments about the neutral axis gives

$$M = E I_{zz} \kappa \implies \kappa = \frac{M}{E I_{zz}} = \frac{\pi P R^2 r^2 r_p}{E I_{zz} (R^2 - r^2)}$$

- 2 M is uniform, so κ is uniform: the centreline is a circular arc of radius $\rho = 1/\kappa$, and the tip turns by (E is the small-strain modulus, $E = 3\mu$)

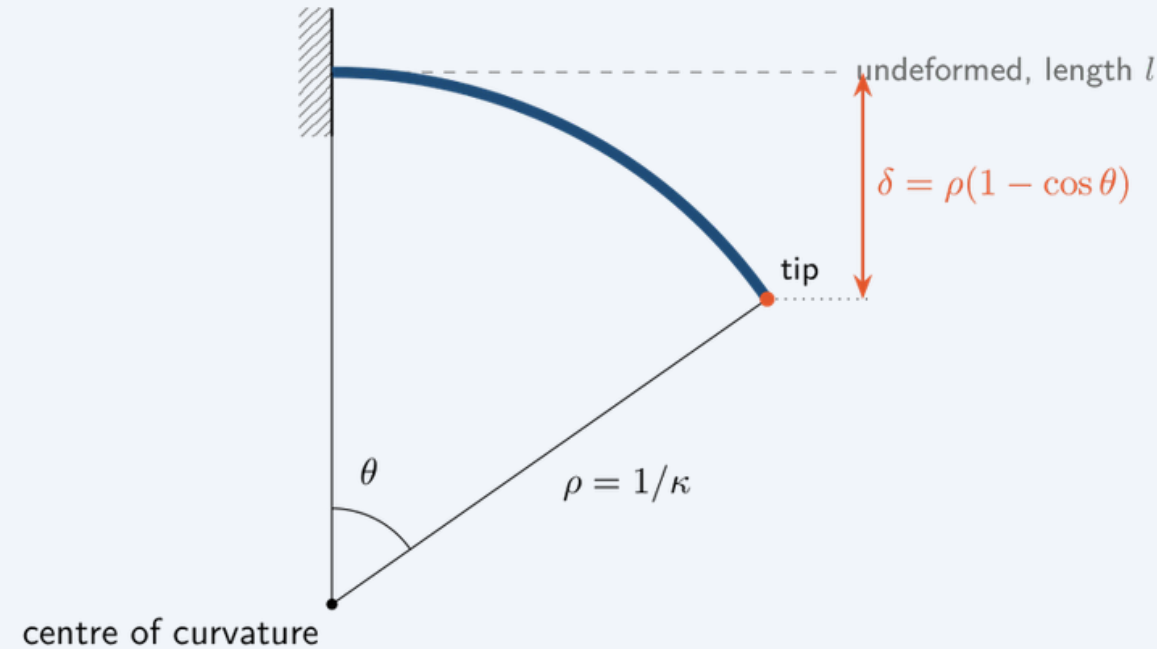
$$\theta = \kappa l = \frac{\pi P R^2 r^2 r_p l}{E I_{zz} (R^2 - r^2)}$$

- 3 Arc geometry (figure): the tip moves sideways by

$$\delta = \rho (1 - \cos \theta) = \frac{1 - \cos \theta}{\kappa}$$

- 4 For small angles $1 - \cos \theta \approx \theta^2/2$, which recovers the cantilever result:

$$\delta \approx \frac{\theta l}{2} = \frac{M l^2}{2 E I_{zz}} \quad (20 \text{ kPa: } \theta = 4.82^\circ, \delta = 4.20 \text{ mm})$$



Uniform moment $\rightarrow$ constant curvature $\rightarrow$ the centreline is a circular arc of radius $\rho = 1/\kappa$ turning through θ .

FINDING

$\theta = 4.82^\circ$, $\delta = 4.20$ mm at 20 kPa, linear in P . Holds only if each fibre is in uniaxial stress, which is true with a hoop sleeve.

The $(1 - 2\nu)$ factor: why a bare actuator barely bends

What we did: put back the in-plane wall stresses that the beam model ignores

HOW — STEP BY STEP

- 1 The beam model ignores the wall stresses *in the section plane* (radial σ_r , hoop σ_θ). Put them back with the full Hooke's law:

$$\varepsilon_x = \frac{\sigma_x - \nu(\sigma_y + \sigma_z)}{E} \implies \sigma_x = E[\varepsilon_0 + \kappa(y - \bar{y})] + \nu(\sigma_y + \sigma_z)$$

- 2 We never need the in-plane field itself. Equilibrium plus the divergence theorem gives its total and where it acts: $2P\pi r^2$, through the cavity centre.

$$\int_A (\sigma_y + \sigma_z) dA = 2P\pi r^2, \quad \int_A (\sigma_y + \sigma_z)(y - \bar{y}) dA = 2P\pi r^2 e$$

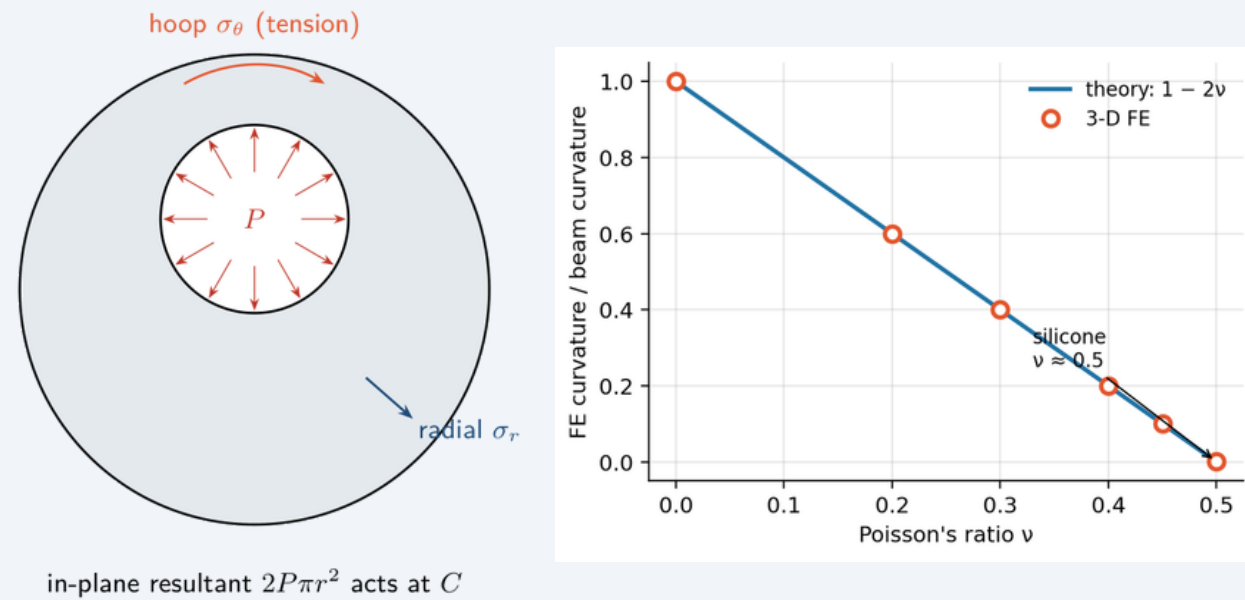
- 3 Force balance on the cut (panel 3, step 4):

$$EA\varepsilon_0 + 2\nu P\pi r^2 = P\pi r^2 \implies \varepsilon_0 = (1 - 2\nu) \frac{P\pi r^2}{EA}$$

- 4 Moment balance on the cut:

$$EI_{zz}\kappa + 2\nu P\pi r^2 e = P\pi r^2 e \implies \kappa = (1 - 2\nu) \frac{M}{EI_{zz}}$$

- 5 Silicone is incompressible, $\nu \rightarrow 0.5$: $\kappa \rightarrow 0$ and $\varepsilon_0 \rightarrow 0$. A bare actuator does **not** bend at first order. A hoop sleeve takes the hoop stress and restores $\kappa = M/(EI_{zz})$.



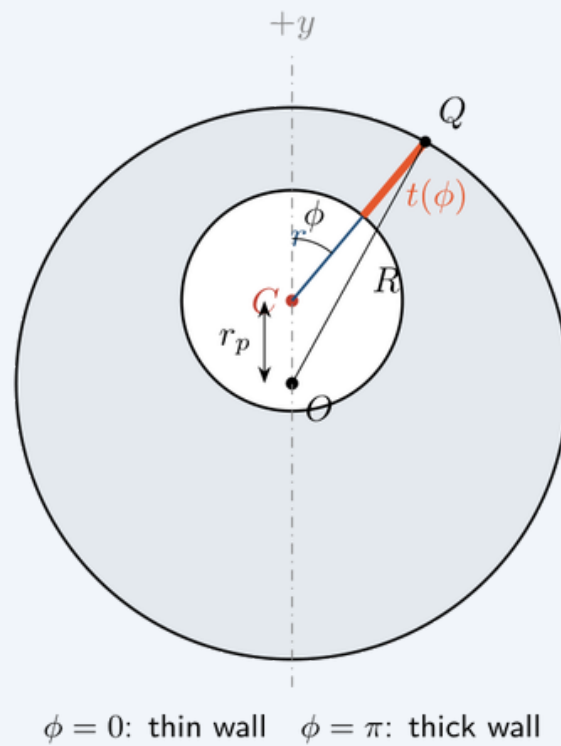
Pressure also loads the wall in its own plane (left). Through Poisson's ratio this cancels the thrust moment; the 3-D FE curvature follows $1 - 2\nu$ exactly (right).

FINDING

$\kappa = (1 - 2\nu) \cdot M / (EI_{zz}) \rightarrow 0$ for silicone. A bare actuator does not bend at first order; the beam formula holds only with a hoop sleeve.

Wall thickness and a linear wall-expansion estimate

What we did: derived the wall thickness $t(\phi)$ and tested a curvature that is linear in the wall expansion



$\phi = 0$: thin wall $\phi = \pi$: thick wall

Wall thickness $t(\phi)$ along a ray from the cavity centre C: 3 mm at $\phi = 0$ (thin wall), 9 mm at $\phi = \pi$ (thick wall).

FINDING

$t(\phi)$ is exact; the FE mesh is built on it. A wall-expansion curvature that is linear in P cannot be added to the beam term.

HOW — STEP BY STEP

- 1 Draw a ray from the cavity centre C at angle ϕ from $+y$. It leaves the outer circle at Q , a distance $s = r + t$ from C . Law of cosines in triangle OCQ :

$$R^2 = r_p^2 + s^2 + 2 r_p s \cos \phi$$

- 2 Solve the quadratic for s (positive root) and subtract r :

$$t(\phi) = \sqrt{R^2 - r_p^2 \sin^2 \phi} - r_p \cos \phi - r \quad t(0) = t_{\min} = 3, \quad t(\pi) = t_{\max} = 9 \text{ mm}$$

- 3 Thin-membrane hoop stress: the thin wall carries three times the hoop stress of the thick wall.

$$\sigma_{\theta}(\phi) \approx \frac{P r}{t(\phi)}$$

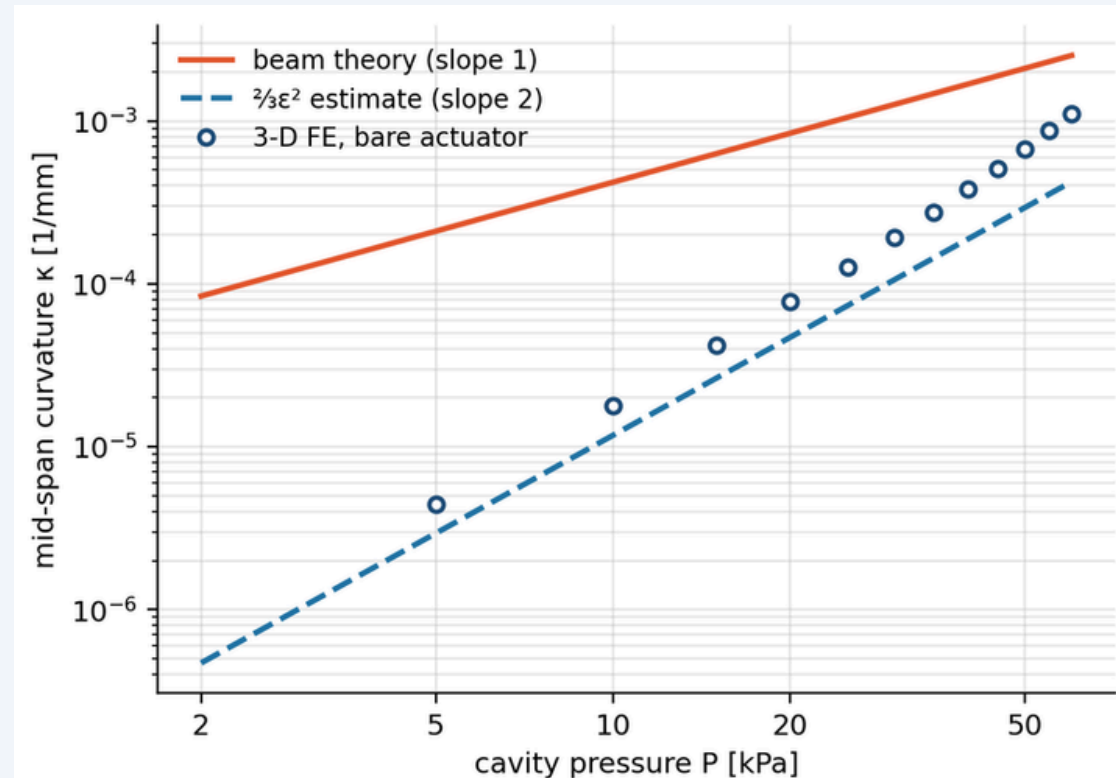
- 4 A first guess takes a *linear* axial strain $\varepsilon_x \propto Pr/(Et)$ and difference it across the section:

$$\kappa_{\text{exp}} \approx \frac{\varepsilon_x(\text{thin}) - \varepsilon_x(\text{thick})}{2R} = \frac{Pr}{2ER} \left(\frac{1}{t_{\min}} - \frac{1}{t_{\max}} \right)$$

- 5 This cannot be added to κ_{thrust} : it uses the same linear Poisson coupling that, for $\nu = 0.5$, *cancels* the thrust moment (previous slide). The real effect is second order (next slide).

What really bends a bare actuator: second-order lengthening

What we did: expanded the large-strain equations of a closed rubber tube to second order



On log axes the FE curvature of the bare actuator rises with slope 2, like the $\frac{2}{3}\epsilon^2$ estimate — not slope 1 like beam theory.

FINDING

Axial strain $\approx \frac{2}{3}(\text{hoop strain})^2$, so curvature grows like P^2 , not P . The thin wall finally balloons at $P_{\text{lim}} = 0.523 \mu = 104.7 \text{ kPa}$.

HOW — STEP BY STEP

- 1 Thin closed neo-Hookean tube. Hoop, axial and thickness stretches $\lambda_1 = 1+a$, $\lambda_2 = 1+b$, $\lambda_3 = 1/(\lambda_1\lambda_2)$. Membrane stresses:

$$\sigma_\theta = \mu(\lambda_1^2 - \lambda_3^2), \quad \sigma_x = \mu(\lambda_2^2 - \lambda_3^2)$$

- 2 A closed end carries half the hoop load axially, $\sigma_x = \sigma_\theta/2$:

$$2\lambda_2^2 = \lambda_1^2 + \lambda_3^2, \quad \lambda_3^2 \approx 1 - 2a - 2b + 3a^2 + 3b^2 + 4ab$$

- 3 First order in a : $4b = -2b$, so $b = 0$. No linear lengthening, the same result as the $(1-2\nu)$ slide.

- 4 Second order, with $b = O(a^2)$: $6b = 4a^2$, so

$$\epsilon_{\text{axial}} \approx \frac{2}{3} \epsilon_{\text{hoop}}^2$$

- 5 The thin wall has the larger hoop strain, so it lengthens more and the actuator curls toward the thick wall, with $\kappa \propto P^2$ at low pressure (FE: slope 2). Past a limit the wall thins faster than its stress can rise:

$$P_{\text{lim}} = 0.523 \mu \quad (104.7 \text{ kPa for } E = 0.6 \text{ MPa})$$

The derivation at a glance: key results

Nine results carry the whole model. Every step is written out in [report/final/btp_derivation.pdf](#)

#	Quantity	Result	Holds for
1	neutral axis	$\bar{y} = -r^2 r_p / (R^2 - r^2)$	any section
2	second moment	$I_{zz} = \pi(R^4 - r^4)/4 - \pi R^2 r^2 r_p^2 / (R^2 - r^2)$	any section
3	thrust and moment	$F = P\pi r^2, \quad M = \pi P R^2 r^2 r_p / (R^2 - r^2)$	any material
4	tip angle	$\theta = \kappa l = M l / (EI_{zz})$	hoop-reinforced
5	tip deflection	$\delta = (1 - \cos \theta) / \kappa \approx \theta l / 2$	hoop-reinforced
6	bare curvature	$\kappa = (1 - 2\nu) M / (EI_{zz}) \approx 0$	bare, $\nu \rightarrow 0.5$
7	wall thickness	$t(\phi) = \sqrt{R^2 - r_p^2 \sin^2 \phi} - r_p \cos \phi - r$	exact
8	wall lengthening	$\varepsilon_{ax} \approx \frac{2}{3} \varepsilon_{hoop}^2$, so $\kappa \propto P^2$	bare, low P
9	ballooning limit	$P_{lim} = 0.523 \mu$ (104.7 kPa for $E = 0.6$ MPa)	bare, neo-Hookean

Four model tiers, from closed form to 3-D FE

What we did: built and compared four models of increasing fidelity (actuator_model.py, actuator_fea.py)

Tier 1 · Linear beam

Closed-form Euler–Bernoulli result.
Instant; valid for the sleeved actuator at low pressure.

Tier 2 · Arc kinematics

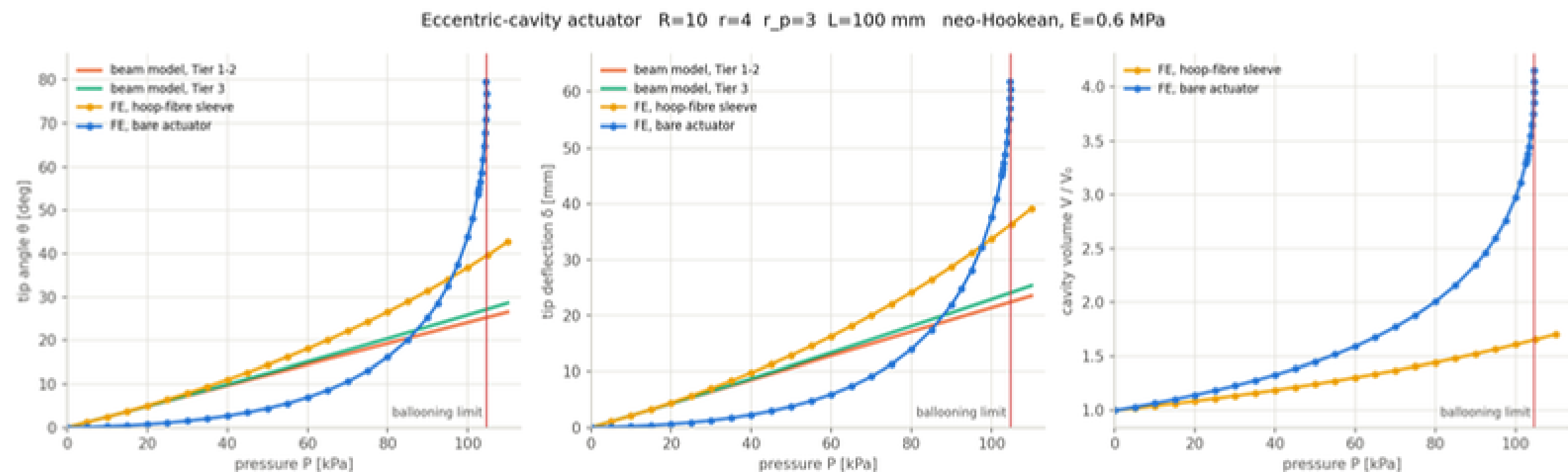
Tier 1 curvature with large rotation and axial stretch: $x_{\text{tip}} = \rho \sin \theta$, $\delta = \rho(1 - \cos \theta)$.

Tier 3 · Hyperelastic section

Every fibre follows a Yeoh / neo-Hookean law; section equilibrium solved numerically.

Tier 4 · Nonlinear 3-D FE

Full large-strain solid model; the only tier that captures the bare actuator.



FINDING

Tiers 1–3 follow the sleeved FE within 5 % up to $P/E \approx 0.03$ (FE is 25 % higher at $P/E = 0.1$). No beam tier fits the bare actuator: FE is 4–10× lower at low pressure, then runs away near 104.7 kPa.

HOW — same geometry (R 10, r 4, r_p 3, L 100 mm) and silicone (E = 0.6 MPa) in every tier; beam tiers plotted against the sleeved and bare FE sweeps (fea_compare.py).

FEA process: an in-house nonlinear 3-D solver

What we did: wrote and verified our own large-deformation FE code in Python (no FE package was available)

- 1 · Mesh**

Rays from the cavity centre using $t(\phi)$; 588 hex27, half model
- 2 · Material**

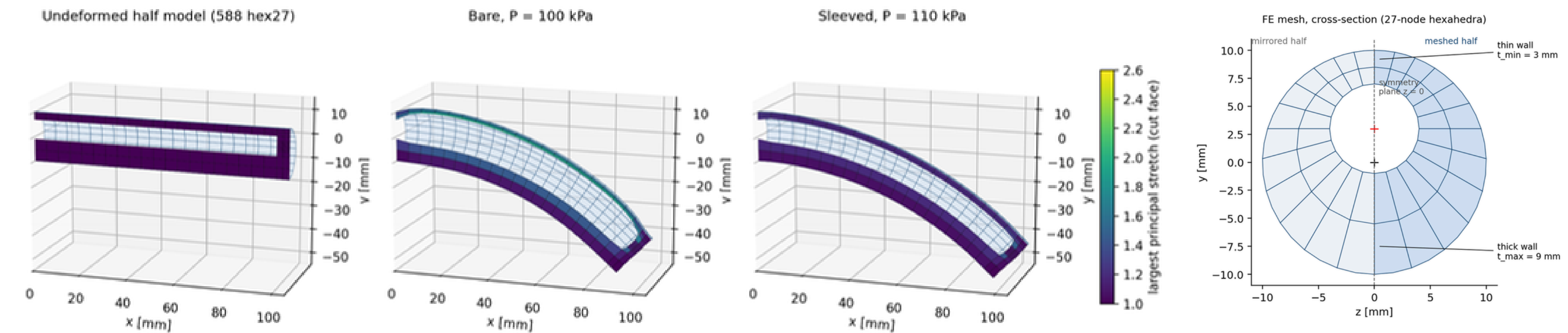
Neo-Hookean / Yeoh strain energy $W(F)$ for silicone
- 3 · Incompressible**

Mixed displacement–pressure (Q2/P1) elements, $\nu = 0.4995$
- 4 · Load & supports**

Follower pressure from $-P \cdot V$; clamped base, symmetry plane
- 5 · Solve**

Newton–Raphson; tangent from JAX automatic differentiation
- 6 · Follow the path**

Pressure steps, then volume control over the ballooning peak



FINDING

The mesh follows the exact wall thickness $t(\phi)$ from the derivation. One pressure sweep takes 2–3 minutes on a laptop (36 states bare, 23 sleeved). Every step of the method is explained in [report/fea_handbook.pdf](#).

How the solver finds equilibrium

How: minimise the total energy with Newton iterations, and follow the path past the pressure peak

1 Unknowns

Displacements of 6,205 nodes, plus 4 pressure values per element that hold the silicone nearly incompressible ($\nu = 0.4995$).

2 Energy

$\Pi(u) = \sum \int W(F) dV - P \cdot V_{\text{cavity}}$. Equilibrium is where Π is stationary: $R(u) = \partial \Pi / \partial u = 0$.

3 Newton–Raphson

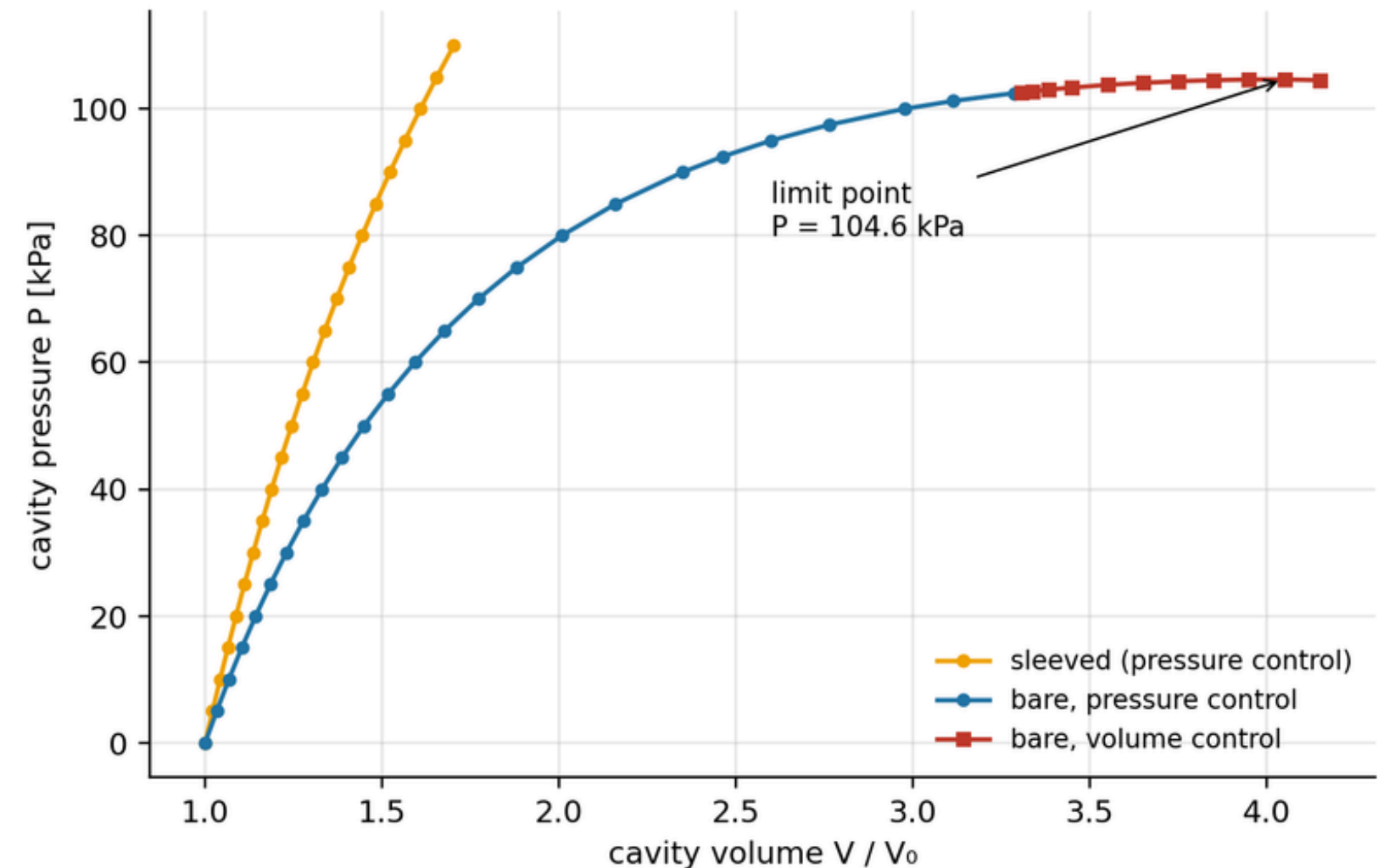
Repeat $K \cdot \Delta u = -R$ until $R \approx 0$. The tangent $K = \partial^2 \Pi / \partial u^2$ comes from JAX automatic differentiation — no hand-derived stiffness.

4 Follower pressure

Pressure acts on the deformed cavity; its nodal forces are $P \cdot \partial V / \partial u$, so they turn with the wall automatically.

5 Path following

Pressure steps up to the peak; there K turns singular, so we prescribe the cavity volume and solve for P instead.

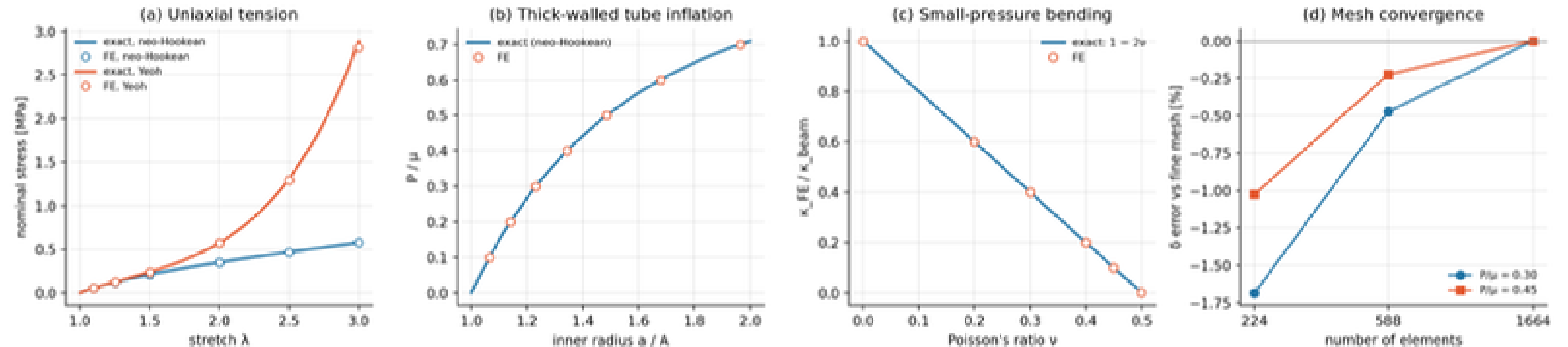


FINDING

Pressure control alone stops at the peak. Volume control — like driving the actuator with a syringe — follows the bare actuator over its limit at 104.7 kPa.

Verifying the solver against exact solutions

What we did: four benchmark tests with known answers before trusting any actuator result (fea_verify.py)



0.2 %

max error against the exact stress,
stretch up to 3

0.05 %

max error against exact tube inflation, P/μ
up to 0.7

0.0002

max deviation from the $1 - 2\nu$ law, $\nu = 0$
to 0.4995

0.5 %

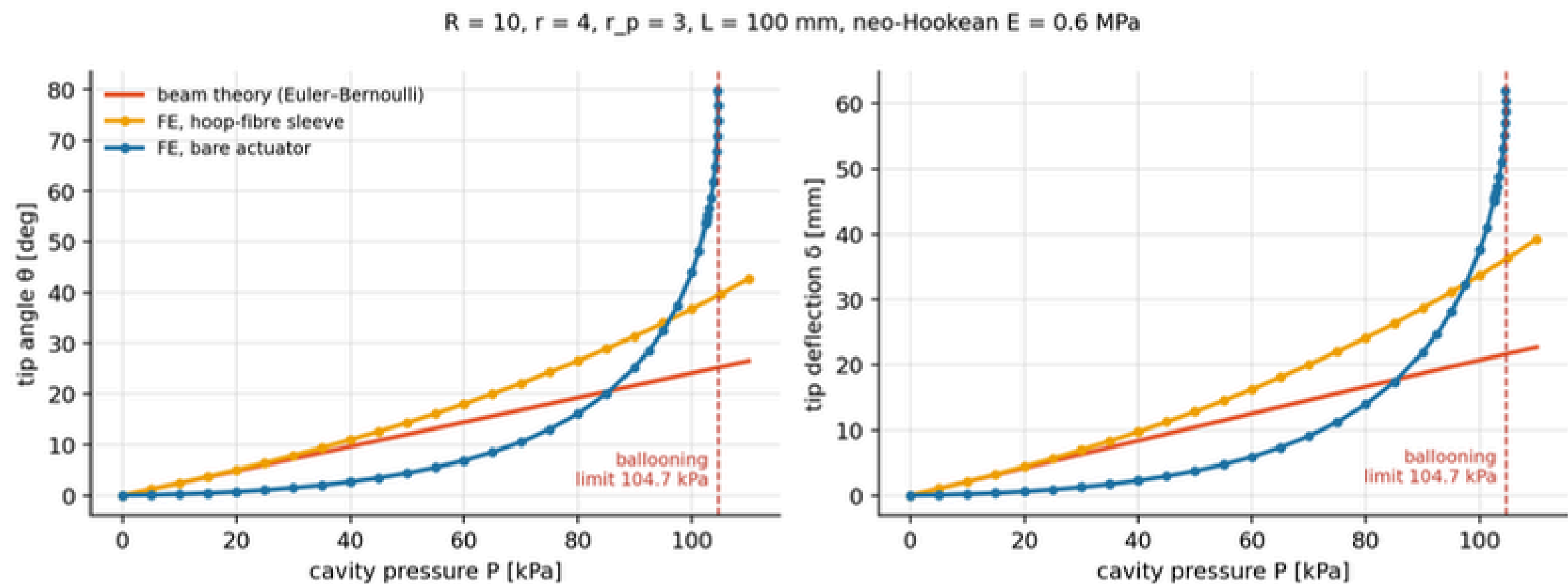
medium mesh (588) vs fine mesh (1664)
in tip deflection

FINDING

The code reproduces exact large-strain solutions within a fraction of a percent and the medium mesh is converged. The FE results can be trusted; the main remaining uncertainty is the silicone's material data, which only tests can fix.

Bare vs sleeved: two very different responses

What we did: pressure sweeps 0–110 kPa on both actuators, compared with beam theory (same geometry and silicone)



$\leq 5 \%$

sleeved FE vs beam theory, up to $P/E \approx 0.03$

4–10×

beam theory over-predicts the bare actuator at low pressure

104.7 kPa

bare ballooning limit = 0.523μ ($E = 0.6 \text{ MPa}$)

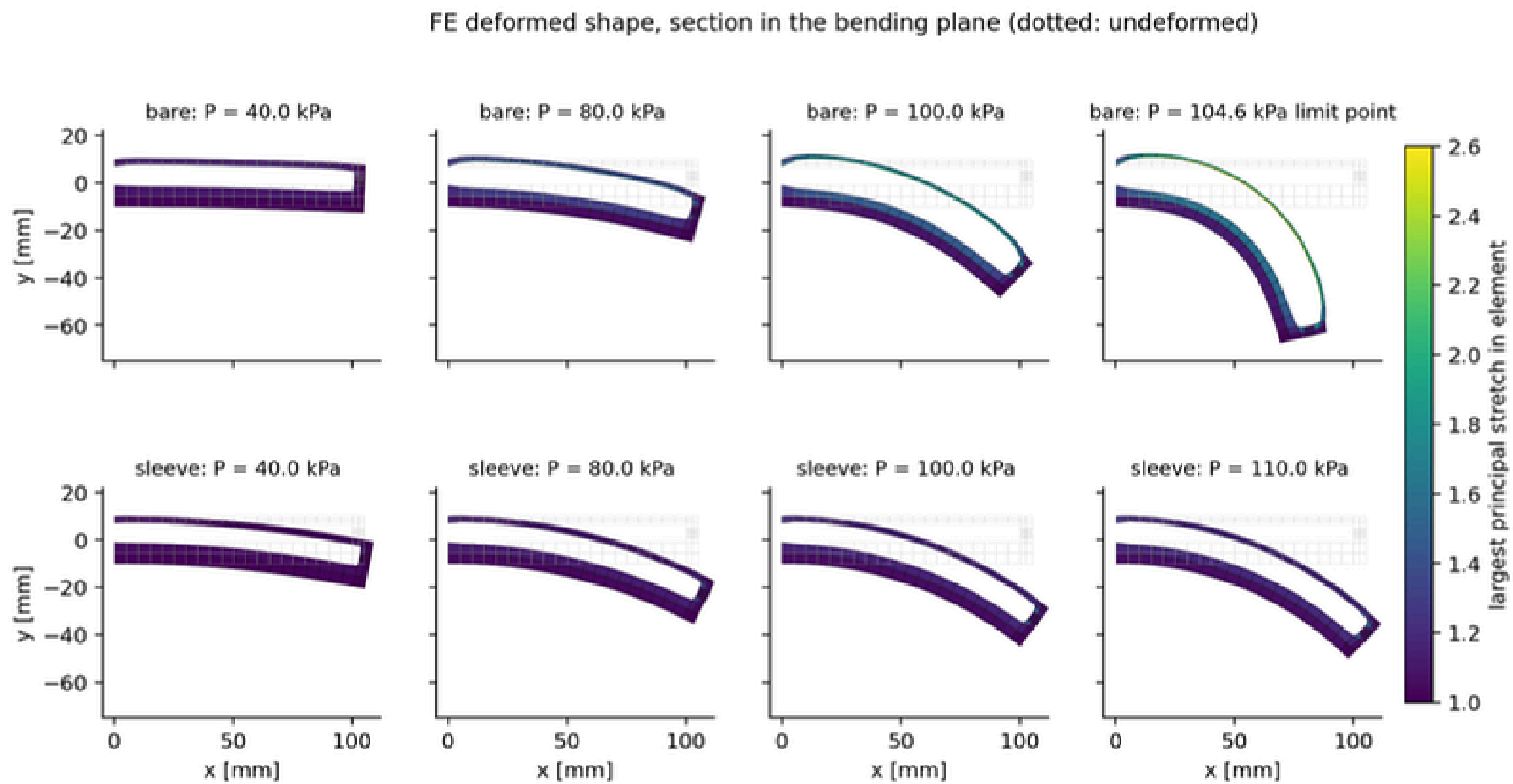
Tip deflection δ : 60 kPa → bare 5.9 mm, sleeved 16.3 mm · 100 kPa → bare 37.6 mm, sleeved 33.7 mm · at the limit → bare 60 mm

FINDING

The sleeve makes the response near-linear and predictable. The bare actuator barely moves at low pressure, then runs away near its ballooning limit.

Deformed shapes: where the bare actuator balloons

What we did: cut the deformed model along the bending plane at 40, 80, 100 kPa and at the limit; colour = stretch



BARE (TOP ROW)

The thin wall balloons. At the 104.7 kPa limit the cavity holds 4× its volume and the thin wall is ⅓ of its thickness.

SLEEVED (BOTTOM ROW)

Smooth, uniform curvature; no ballooning up to 110 kPa.

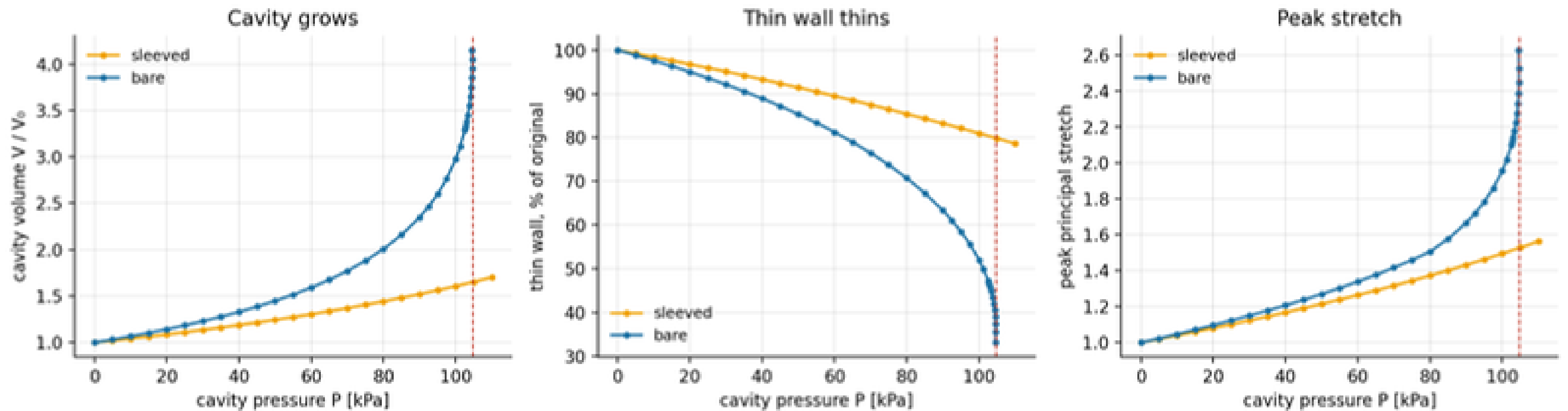
FINDING

Reinforce the actuator for a predictable $P \rightarrow \delta$ map. Run a bare actuator below about 90 % of its measured limit pressure.

Dotted outline = undeformed shape. Peak principal stretch reaches 2.5 at the limit.

Inside the wall: volume, thinning and stretch

What we did: tracked cavity volume, thin-wall thickness and peak stretch through each pressure sweep



× 4.05

cavity volume at the bare limit (sleeved: × 1.70 at 110 kPa)

1/3

of the 3 mm thin wall left at the limit (sleeved: 79 % at 110 kPa)

2.5

peak principal stretch at the limit (sleeved: 1.56 at 110 kPa)

FINDING

The thin wall is the weak point: near the limit it thins to a third and stretches 2.5×, so silicone fatigue must also set the operating window. The sleeve keeps all three quantities moderate.

Bare vs reinforced: which actuator to build?

What we did: ran the same pressure sweep on both designs and compared them on six criteria

	BARE (NO REINFORCEMENT)	SLEEVED (HOOP-REINFORCED)
Tip deflection at 60 kPa	5.9 mm	16.3 mm (2.8× more)
Tip deflection at 100 kPa	37.6 mm, then snap-through at 104.7 kPa	33.7 mm, smooth and still rising
Deformed shape	Local balloon on the thin wall	Uniform circular arc
Usable pressure window	About 0.3–0.9 of the limit pressure	From zero to at least 110 kPa
Analytical model	Needs the full nonlinear FE	Beam tiers: 5 % at low P, 20–25 % low at 60 kPa
Fabrication	Single silicone cast	Extra step: thread winding or fibre sleeve

RECOMMENDATION

Build the hoop-reinforced (sleeved) actuator: it bends 2.8× more at moderate pressure, stays predictable across the range and is captured by the simple beam model. Keep one bare specimen to measure the ballooning limit.

Timeline: 9 October – 8 November 2026

What we will do: multi-cavity designs that bend in any x–y direction, an ANSYS check of every model, then mould and experiments



SUMMARY

What we know so far

01

Derivation, step by step

Section, loading and beam response follow from statics. The $(1-2\nu)$ factor shows that axial stretch alone barely bends a silicone actuator; the bending comes from the wall ballooning.

02

Three analytical model tiers

Linear beam $\rightarrow$ $(1-2\nu)$ factor $\rightarrow$ wall expansion. Valid only for the reinforced actuator: within 5 % of the FE at low pressure, 20–25 % low at 60 kPa.

03

Nonlinear 3-D FE, written and verified

27-node mixed u–p hexes, follower pressure and Newton with autodiff tangents. Four benchmark checks agree to within 0.5 %.

04

Build the reinforced actuator

The bare one balloons and snaps through at 104.7 kPa. The sleeved one bends 2.8× more at 60 kPa and stays smooth and predictable.

Next: multi-cavity designs $\rightarrow$ ANSYS check $\rightarrow$ mould and experiments from 26 Oct

Thank you — questions?